THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE TOWN COUNCIL

FORM FOUR PRE- MOCK EXAMINATION APRIL -2024

031/2 **PHYSICS 2**
(ACTUAL PRACTICAL)

TIME: 2:30 HOURS **24TH APRIL 2024**

INSTRUCTIONS

1. This paper consists of two questions. Answer all questions
2. Each question carries 25 marks.
3. All writings should be in blue/black ink EXCEPT drawings if any which must be in pencil.
4. Non-programmable calculators and mathematical tables may be used.
5. Write your name on every page of your answer sheet.
6. Write your answers in the answer sheet provided
7. The following information may be useful:
   Pie (π) = 3.14

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1. The aim of this experiment is to deduce a value of length fixed on a simple pendulum system.
You are provided with a retort stand with its accessories, two cork pads, a pendulum bob and a cotton thread of about 100cm.
Proceed as follows:

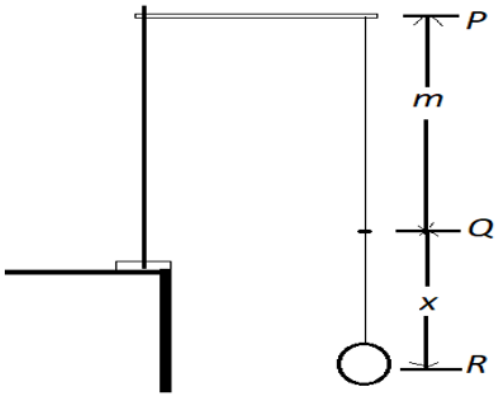


Figure 1

(a) Tie a knot, Q such that x is at a distance of 30cm. Suspend the pendulum as shown in Figure 1 above and m, is the distance between the point of suspension P and Q on the string.
(b) Adjust the string at its point of suspension so that m= 60 cm. Swing the pendulum and measure the time, t for 10 oscillations of small amplitude. Calculate the periodic time, T and hence determine, T^2.
(c) Record the values of m, t, T, and T^2 in a tabular form.
(d) Repeat the above procedures for values of m = 50 cm, 40 cm, 30 cm and 20 cm.
(e) Plot a graph of T^2 against m.
(f) From the graph:
  (i) Determine the values of the slope and the T^2-intercept of the graph.
  (ii) Given that $T=2\pi\sqrt{\frac{(m+x)}{g}}$, use the answer obtained in (f) (i) above to compute the value of x and the acceleration due to gravity, g.
  (iii) What is the physical significance of x.?
(g) State any three sources of error. **(25 MARKS)**

2. The aim of this experiment is to determine the relative refractive index for light passing from air to glass.
You are provided with a rectangular glass block, four optical pins, plane papers and a soft board.

Proceed as follows:

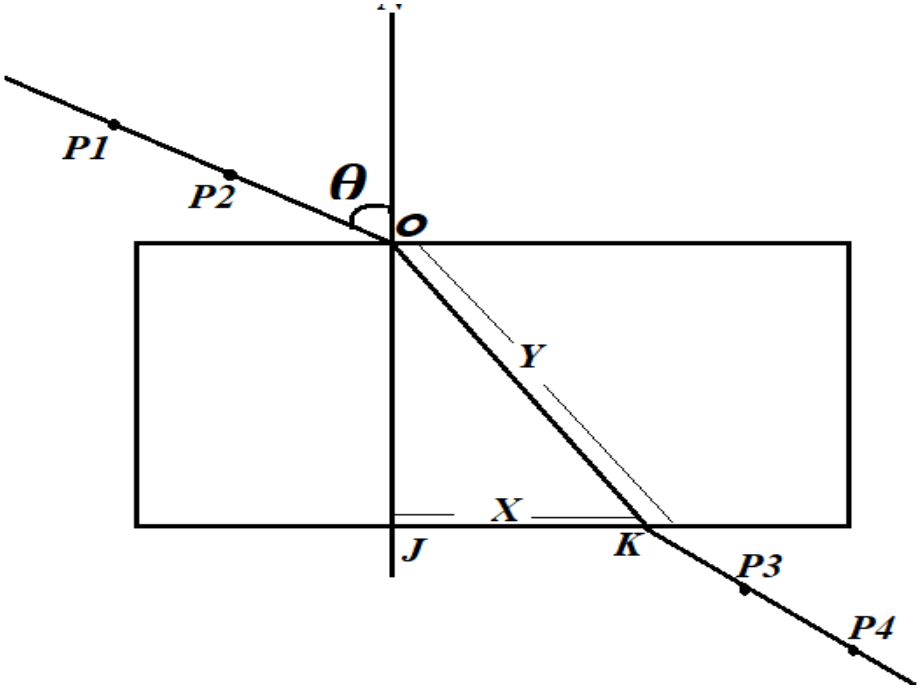


Figure 2

(a) Place a rectangular glass block flat on a white sheet of paper and use pencil to trace the outline of the glass block. Remove the glass block and draw the line with θ of 35^0 with the normal **ON** and erect two pins **P1** and **P2** at suitable positions as shown in figure 2 above.

(b) Replace the block and erect other two pins **P3** and **P4** at positions which appear to be in a straight line with the other two seen through the glass block. Again remove the block and draw complete path of the ray.

(c) Measure the distance **X (**from **J** to **K)** and distance **Y (**from **O** to **K)**

(d) Repeat the procedures (a)-(c) for θ $=45^0$, 55^0, 65^0 and 75^0 and in each case make your drawing on a fresh part of the drawing paper. Record clearly the values of $\boldsymbol{\theta}, \boldsymbol{X}$ **and** $\boldsymbol{Y}$.

(e) Tabulate your results for the values of $\boldsymbol{\theta}$**, X, Y** and including sin $\boldsymbol{\theta}$ *and* $\mu = \frac{\boldsymbol{X}}{\boldsymbol{Y}}$

(h) Plot a graph of sin $\boldsymbol{\theta}$ against **μ**

(i) Find the gradient of your graph.

(j) (i) Given that $\frac{\mathbf{1}}{\boldsymbol{M}}$ **sin** $\boldsymbol{\theta}$ **= μ,** use this equation to find the value of **M**

(ii) What is the physical meaning of **M?**

(k) State three possible sources of error. **(25 MARKS)**

NB: every drawing of the glass block should be attached together with the answer sheets.